

Module 2 - JavaScript Features & Characteristics

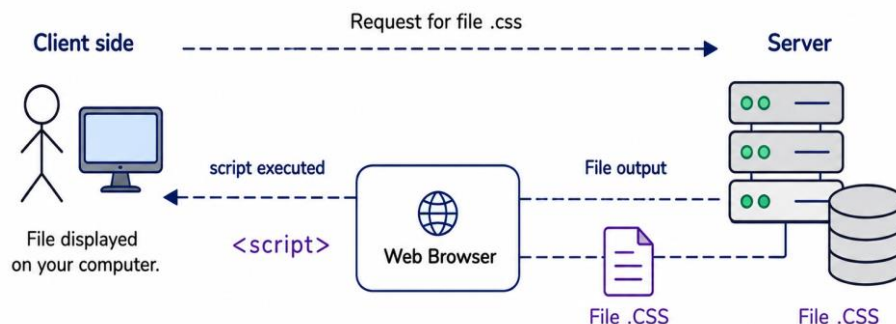
1. What are the features and characteristics of JavaScript?

- JS is mainly used for Client-side scripting
- Interpreted programming language
- Dynamically typed language
- Asynchronous programming language
- Event-driven programming Model
- Single Page Applications (SPA)
- Object-based scripting language
- Platform independent

2. Client-Side Scripting

JavaScript code is executed directly in the user's web browser to make web pages interactive and dynamic.

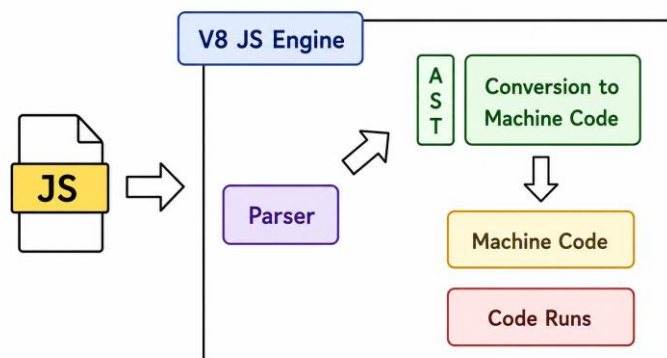
Client-Side Scripting Process



3. Interpreted Programming Language

JavaScript code is executed line by line by the V8 Engine or JavaScript engine.

Interpreted Programming Language (JavaScript)

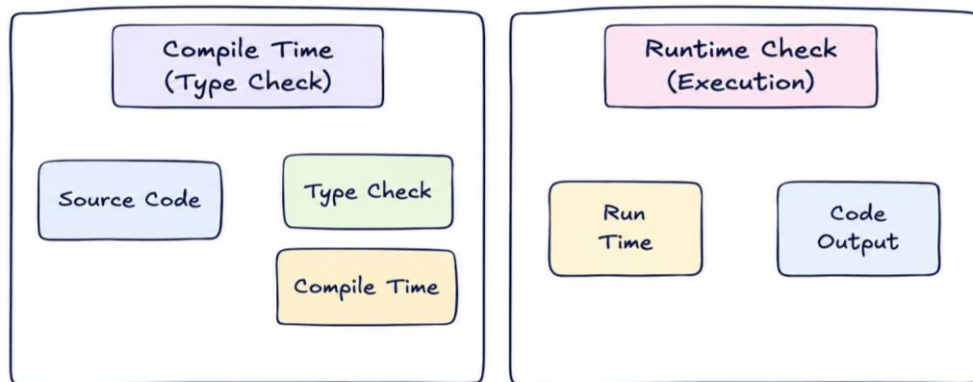


Note:

- In **Client-Side**, JS runs in the browser using the **V8 engine**.
- In **Server-Side**, JS runs using the **Node.js runtime environment** with the **V8 engine**.

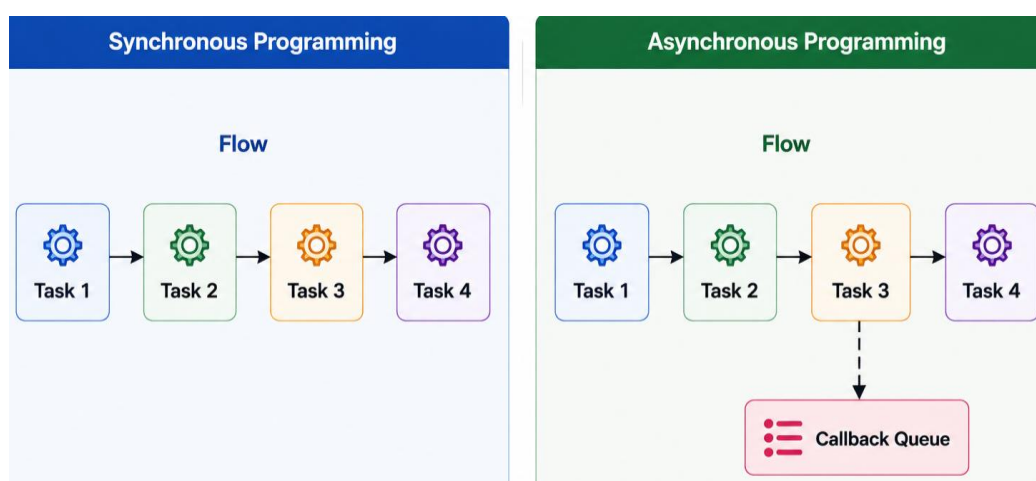
4. Statically Typed vs Dynamically Typed

Statically Typed	Dynamically Typed
Type is checked during compile time.	Type is checked during runtime.
Examples: Java, C++	Examples: JavaScript, Python



5. Synchronous vs Asynchronous Programming

Synchronous Programming	Asynchronous Programming
Tasks are executed one after another.	Tasks can run without waiting for another task to finish.
The next task waits until the current task finishes.	Non-blocking execution.
Blocking execution.	Improves performance.

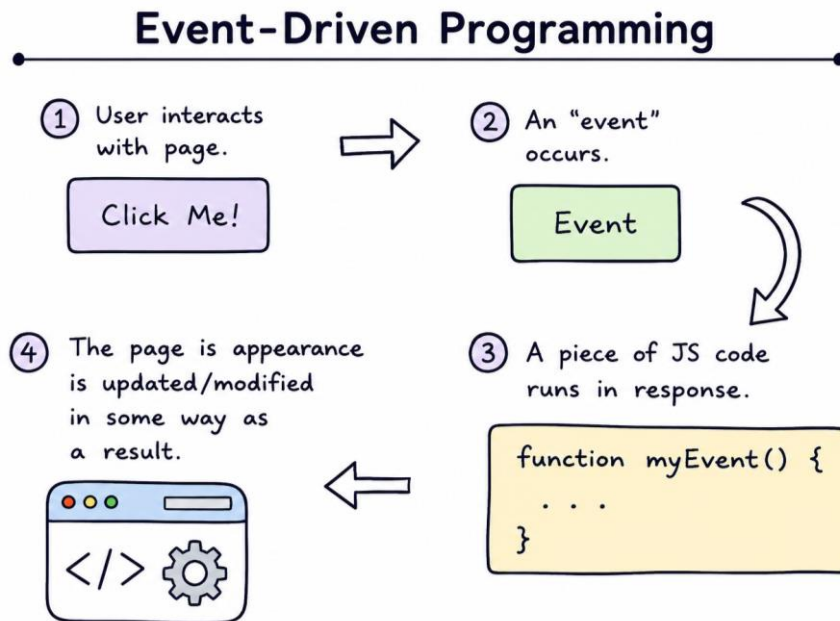


Methods of Asynchronous Programming:

- Callbacks
- Callback Hell
- Promises
- Async / Await

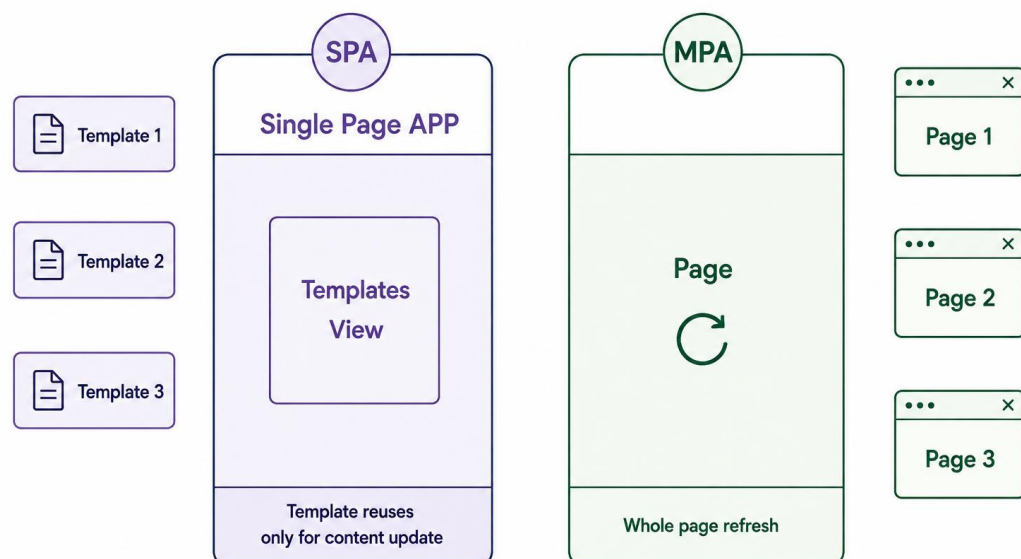
6. Event-Driven Programming

JavaScript responds to user actions or events such as click, keypress, mouseover, input, load, scroll and submit.



7. Single Page Application vs Multi Page Application

Single Page Application (SPA)	Multi Page Application (MPA)
Only the required content is updated.	The entire page reloads on every request.
No full page reload.	Each page is loaded separately from the server.
Provides a faster user experience.	Slower compared to SPA.
Content is dynamically loaded without loading a new HTML page.	Multiple HTML pages are used for different sections.



Examples of SPA frameworks: React, Angular, Vue.js.

8. Object-Based Scripting Language

JavaScript uses objects to organize and manage related data and functionality.

9. Platform Independent

JavaScript can run on different operating systems and devices through a compatible web browser or runtime environment.

10. Why is JavaScript called a High-Level Scripting Language?

JavaScript code is easy to write, read, and understand without dealing directly with low-level system details.

11. What are the Applications of JavaScript?

- **Frontend** – React, Angular, Vue.js
- **Game Development** – Babylon.js, Three.js, Phaser.js
- **Machine Learning** – TensorFlow.js
- **IoT** – Johnny-Five
- **Backend** – Node.js
- **Desktop App Development** – Electron
- **Mobile App Development:**
 - **Native Apps** – React Native
 - **Hybrid Apps** – Ionic, Cordova